Power Budget

Team Number:	TBD
Project Name:	**Rover Environmental Sensor PCB**
Team Member Names:	**TBD**
Version:	**v1.0 (2026-03-10)**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements							
All Major Components	**Component Name**	**Part Number**	**upplyVoltageRange**	**#**	**uteMaximumCurrent (n**	**alCurrent(mA)**	**Unit**
	PIC18F57Q43 Microcontroller	PIC18F57Q43-I/PT	1.8V - 5.5V	1	200.00	200.00	mA
	SHTC3 Humidity Sensor	SHTC3-TR-10KS	2.7V - 5.5V	1	0.40	0.40	mA
						0	mA
						0	mA
						0	mA
						0	mA
B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.							
+12V Power Rail (unused)	**Component Name**	**Part Number**	**upplyVoltageRange**	**#**	**uteMaximumCurrent (n**	**alCurrent(mA)**	**Unit**
						0	mA
						0	mA
						0	mA
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	0	mA
c1. Regulator or Source Ch	N/A	N/A	N/A	1	0	0	mA
					Total Remaining Current Available on +12V Rail	0	mA
+5V Power Rail (unused)	**Component Name**	**Part Number**	**upplyVoltageRange**	**#**	**uteMaximumCurrent (n**	**alCurrent(mA)**	**Unit**
						0	mA
						0	mA
						0	mA
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	0	mA
c2. Regulator or Source Ch	N/A	N/A	N/A	1	0	0	mA
					Total Remaining Current Available on +5V Rail	0	mA
-5V Power Rail (unused)	**Component Name**	**Part Number**	**upplyVoltageRange**	**#**	**uteMaximumCurrent (n**	**alCurrent(mA)**	**Unit**
						0	mA
						0	mA
						0	mA
						0	mA

					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on -5V Rail	0	mA
c3. Regulator or Source Ch	N/A	N/A	N/A	1	0	0	mA
					Total Remaining Current Available on -5V Rail	0	mA
+3.3V Power Rail	**Component Name**	**Part Number**	**upplyVoltageRange**	**#**	**uteMaximumCurrent (n**	**talCurrent(mA)**	**Unit**
	PIC18F57Q43 Microcontroller	PIC18F57Q43-I/PT	1.8V - 5.5V	1	200.00	200.00	mA
	SHTC3 Humidity Sensor	SHTC3-TR-10KS	2.7V - 5.5V	1	0.40	0.40	mA
						0	mA
						0	mA
					Subtotal	200.4	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	250.5	mA
c4. Regulator or Source Ch	+3.3V buck regulator	LM2595S-3.3/NOPB	4.5V - 40V	1	1000.00	1000.00	mA
					Total Remaining Current Available on 3.3V Rail	749.5	mA
C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining							
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails							
External Power Source 1 (s	**Component Name**	**Part Number**	**upplyVoltageRange**	**Output**	**uteMaximumCurrent (n**	**talCurrent(mA)**	**Unit**
Power Source 1 Selection	12V Wall Supply	SWI24-12-N-P5	90VAC - 264VAC	+12V	2000.00	2000.00	mA
Power Rails Connected to External Power Source 1	+3.3V buck regulator	LM2595S-3.3/NOPB	4.5V - 40V	1	1000	1000	mA
						0	mA
						0	mA
					Total Remaining Current Available on External Power Source 1	1000	mA
External Power Source 2 (u	**Component Name**	**Part Number**	**upplyVoltageRange**	**Output**	**uteMaximumCurrent (n**	**talCurrent(mA)**	**Unit**
Power Source 2 Selection						100	mA
Power Rails Connected to External Power Source 2						0	mA
						0	mA
						0	mA
					Total Remaining Current Available on External Power Source 2	100	mA
E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in							
	Component Name	**Part Number**	**upplyVoltageRange**		**Capacity(mAh)**	**iredByRegulators**	

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit